Results

Baseline versus both GenAI combined. We first estimate whether—relative to the *Human only* baseline condition—the GenAI conditions combined causally affect evaluators' third-party assessments and writers' self-assessments of the stories' creativity, in terms of two commonly used dimensions, novelty and usefulness (*12,18*).

We find that evaluators assess that stories composed by writers in the two GenAI conditions are more creative, in terms of both novelty and usefulness (Figure 2A; Figure S1 shows violin plot of raw data). We find that the provision of AI ideas improves the story's novelty by 6.7% (b=0.259, p=0.001, see Table S3; compared to the *Human only* mean of 3.85) and its usefulness by 6.4% (b=0.319, p<0.001; compared to the *Human only* mean of 5.02). Results for each of the constituent components of the novelty index (whether the story is novel, original, and rare) and usefulness index (whether the story is appropriate, feasible, and publishable) are consistent with the aggregate results for the indices (see Table S3).

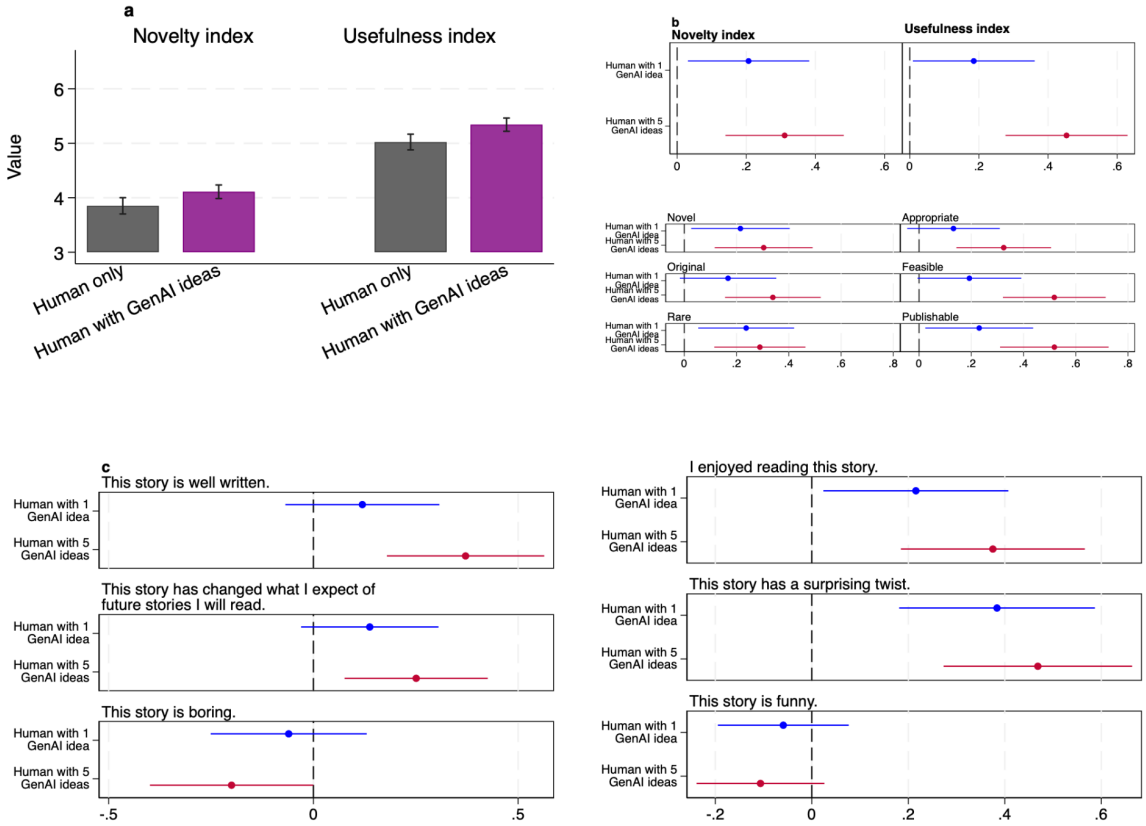

Figure 2. Evaluation of creativity and emotional characteristics by third party evaluators. **a**, Compares novelty and usefulness indices of participants in *Human only* condition to participants who had access to (1 or 5) GenAI ideas combined. **b**, Constituent components of each index. **c–d**, Compares novelty and usefulness indices (and constituent components) (**c**) and emotional characteristics (**d**) of *Human only* condition (dashed vertical line) to *Humans with 1 GenAI idea* and *Humans with 5 GenAI ideas* conditions.

Variation by GenAI conditions. To better understand how greater availability of GenAI ideas affects the enhancement in creativity, we follow our pre-registration to estimate the causal impact of the two GenAI conditions separately. Writers in the *Human with 1 GenAI idea* condition are given the choice to request a single GenAI story idea, while writers in the *Human with 5 GenAI ideas* condition are given the option to access up to five GenAI story ideas.

Across the two GenAI conditions, 88.4% participants choose to call upon GenAI at least once to provide an initial story idea. Of the 100 writers in the *Human with 1 GenAI idea* condition, 82 opted to generate one, while 93 out of 98 writers in the *Human with 5 GenAI Ideas* condition did so. When given the option to call upon GenAI more than once in the *Human with 5 GenAI ideas* condition, participants did so on average 2.55 times, with 24.5% requesting the maximum of five GenAI ideas.

We find that, while having access to one GenAI idea leads to somewhat greater creativity, the most gains (and significant differences in our pre-registered indices) come from writers who have access to five GenAI ideas (Figure 2B; Figure S2 shows violin plot of raw data). With respect to novelty, writers in the *Human with 1 GenAI idea* condition experienced an increase of 5.4% (b=0.207, p=0.021, see Table S4) over writers without GenAI access; whereas writers in the *Human with 5 GenAI ideas* condition showed an increase in novelty of 8.1% (b=0.311, p<0.001) over writers without GenAI access.

The results of story usefulness are even more striking. The usefulness of stories from writers with access to one GenAI idea is 3.7% (b=0.185, p=0.039) higher than that of writers with no GenAI access. Having access to up to five AI ideas increased usefulness by 9.0% (b=0.453, p<0.001) over those with no GenAI access and 5.1% (p=0.0012, compared to the *Human with 1 GenAI idea* mean of 5.21) over those with access to one GenAI idea. The overall results suggest that having access to more AI ideas lead to more creative storytelling. The novelty and creativity index results are qualitatively unchanged when we include evaluator fixed effects, story order fixed effects, story topic fixed effects, and an indicator variable that equals one if the writer accessed at least one GenAI idea (see Table S5).

In contrast, writers self-assessing their own stories showed no significant differences in the novelty and usefulness between authors who were offered GenAI ideas and those who were not (see Tables S6).

Exploratory analyses: emotional characteristics. Next, we turn to measures that gauge the evaluators' emotional responses to the stories, based on categories of general reader interest, including how well written, enjoyable, funny, and boring the stories are and the extent to which the story has a plot twist. We also ask whether the story changed the reader's expectations about future stories (based on literature theorist Robert Jauss' conception of more novel literature changing the reader's "horizon of expectations" in the future (*19*)).

As illustrated in Figure 2C, we find that stories written by writers with access to GenAI ideas are more enjoyable (*Human with 1 GenAI idea*: b=0.216, p=0.028; *Human with 5 GenAI ideas*: b=0.375, p<0.001, see Table S7) and are more likely to have plot twists (*Human with 1 GenAI idea*: b=0.384, p<0.001; *Human with 5 GenAI ideas*: b=0.468, p<0.001). Relative to *Human only* stories, when the writer has access to up to five GenAI ideas, the stories are considered to be better written (b=0.372, p<0.001), have more of an effect on the evaluator's expectations of future stories (b=0.251, p=0.005), and be less boring (b= –0.200, p=0.049). Stories in the *Human with 5 GenAI ideas* are, however, not evaluated as more funny than *Human only*; if anything the coefficient is negative but not significant (b= –0.106, p=0.115).

Again, writers' self-assessment of their own stories showed no significant differences in the story characteristics across conditions (see Tables S8).

Heterogeneity by inherent creativity. Since our human writers were not specifically selected for their creative predispositions or work in creative industries, we are able to take advantage of natural variation in the underlying creativity of writers in our sample. To do so, we had writers complete a Divergent Association Task (DAT) prior to writing their stories[17]. The task entails providing ten words that are as different from each other as possible. The DAT score is the cosine distance of the underlying word embeddings (scaled to 100) and captures the individual's inherent creativity. In our sample, the DAT score had a mean of 77.24 and a standard deviation of 6.48. The computation of DAT requires seven of ten submitted terms to be valid (i.e., single words that appear in the dictionary). Two writers failed to properly submit seven valid words, so the DAT score was successfully computed for 291 of 293 writers.

First, we look at whether different writers engage with GenAI more than others: we do not find differences between more creative writers and less creative writers in how frequently they access GenAI ideas in the two GenAI conditions (see Table S9). Among both more and less creative writers in the *Human with 5 GenAI ideas* condition, all five ideas were requested 24.5% of the time. In short, we do not observe any differences in how GenAI was accessed based on the inherent creativity of the writer.

Next, we interact the continuous DAT score with our conditions (see Tables S10 and S11 for results on all outcome variables). Figure 3 presents graphs that show the differential effect of GenAI ideas on select variables, based on the inherent creativity of the writer (see Figure S3 for graphs of remaining outcome variables). Among the most inherently creative writers (i.e., high-DAT writers), there is little effect of having access to GenAI ideas on the creativity of their stories. Across all conditions, high-DAT writers' stories are evaluated relatively highly, both in terms of novelty and usefulness; and providing them with access to GenAI does not affect their high evaluations. We observe a similar result among high-DAT writers for how well the story was written, how enjoyable and, conversely, how boring it is: having access to GenAI does not affect high-DAT writer's already good performance on these outcomes.

In contrast, access to GenAI ideas substantially improves the creativity and select emotional characteristics of stories written by inherently less creative writers (i.e., low-DAT writers). Among